

# A Fast Algorithm for Generating All Triangulations of Biconnected Plane Graphs

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## Abstract

We present an algorithm for generating all triangulations of a biconnected plane graph  $G$  with  $n$  vertices in  $O(1)$  amortized time per triangulation and  $O(n)$  space. Our algorithm establishes a double-layer tree structure, consisting of a recursive tree structure for each face, and generates triangulations of each face through a genealogical tree structure. The key innovation is a novel “safe root vertex” selection strategy that prevents the multi-edge problem inherent in biconnected graphs, extending the optimal complexity previously achievable only for triconnected graphs. To the best of our knowledge, this is the first algorithm for generating all triangulations of a biconnected plane graph with  $O(1)$  amortized time complexity per triangulation. We present experimental validation on 35 diverse test cases, demonstrating the effectiveness of our approach.

## 1 Introduction

In this paper, we consider the problem of generating all triangulations of biconnected plane graphs, which has many applications in computational geometry [3], VLSI floorplanning [8], and graph drawing [10]. The main challenge in generating all triangulations is the exponential number of possible triangulations. This exponential growth means that if we attempt to store all triangulations, it will require an exponential amount of storage.

There have been two types of algorithms for generating triangulations [1, 12]. One type of algorithm generates all triangulations but outputs only unique triangulations. This approach requires exponential space to store and check triangulations before selecting unique ones. Another type of algorithm generates only canonical triangulations [12], meaning we ensure that each generated triangulation is unique by construction.

There has been substantial research on algorithms for generating triangulations [2, 5, 7, 13]. Li and Nakano [9] provided an algorithm that generates triangulations in  $O(r^2n)$  time per triangulation. Later, Nakano improved the algorithm to achieve  $O(rn)$  time, where  $r$  is the number of outerplanar vertices. Parvez, Rahman, and Nakano [11] provided the most efficient algorithm for generating all triangulations of a triconnected plane graph in  $O(1)$  time per triangulation with  $O(n)$  space complexity.

Table 1 summarizes the key differences between existing algorithms and our contribution. The reverse search approach [1] typically requires  $O(n^2)$  or higher time per solution due to parent identification overhead. While Parvez et al. [11] achieved optimal complexity for triconnected graphs, their approach fails for biconnected graphs due to separation pairs. Our

algorithm extends their optimal complexity to the strictly more general class of biconnected plane graphs.

Table 1: Comparison of triangulation enumeration algorithms

| Algorithm          | Graph Class        | Time/Triang.                       | Space                    |
|--------------------|--------------------|------------------------------------|--------------------------|
| Reverse Search [1] | General            | $O(n^2)$ or worse                  | $O(n)$                   |
| Li-Nakano [9]      | Biconnected        | $O(r^2n)$                          | —                        |
| Nakano (improved)  | Biconnected        | $O(rn)$                            | —                        |
| Parvez et al. [11] | Triconnected       | $O(1)$ amortized                   | $O(n)$                   |
| <b>This work</b>   | <b>Biconnected</b> | <b><math>O(1)</math> amortized</b> | <b><math>O(n)</math></b> |

The algorithm for generating all triangulations of a triconnected plane graph can be applied to biconnected plane graphs as well, but a major problem arises: the “multi-edge” problem. In triconnected graphs [6], there cannot be any separation pairs, meaning there do not exist two vertices whose removal can disconnect the remaining graph.

Biconnected graphs arise naturally in many practical applications, including VLSI layout design where circuit components must remain connected after any single point failure, mesh generation for finite element analysis where the mesh must adapt to domain boundaries that create separation pairs, and network topology design where backup paths are essential. The ability to enumerate all triangulations of biconnected graphs enables exploration of multiple geometric realizations and structural alternatives in these applications.

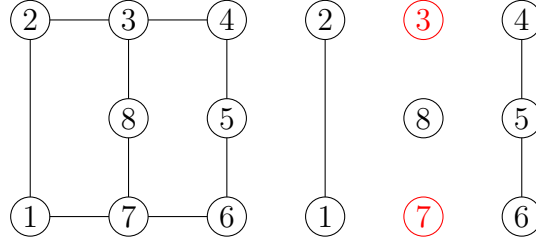


Figure 1: Disconnecting the graph by removing the separation pair (3,7). The remaining graph has three components:  $\{1,2\}$ ,  $\{4,5,6\}$ , and  $\{8\}$ .

In the biconnected graph shown in Figure 1, removing vertices 3 and 7 disconnects the remaining graph. Therefore, (3,7) is a separation pair. However, this property does not occur in triconnected graphs.

In the graph, the (3,7) separation pair can have a chord between them in more than one way.

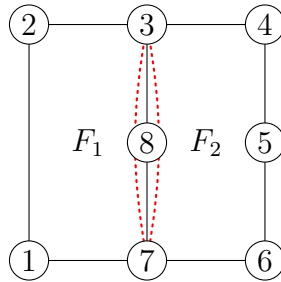


Figure 2: The (3,7) separation pair can have a chord between the vertices in more than one way.

In Figure 2, within this biconnected graph, the (3,7) separation pair can have two chords at the same time: one in face  $F_1$  and another in face  $F_2$ . In the Parvez, Rahman, and Nakano

algorithm [11], we are assured that there will not be any multi-edge problem because multi-edges are not possible in triconnected plane graphs. However, if we want to extend that algorithm to biconnected graphs, there is a significant possibility of multi-edges.

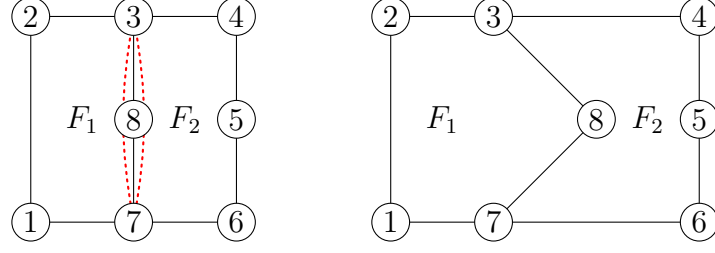


Figure 3: The left graph is biconnected, where  $(3, 7)$  has the possibility of becoming a multi-edge, but in the right graph,  $(3, 7)$  cannot be a multi-edge because it is a triconnected graph.

There cannot be any multi-edge problem in the right graph, as triangulating faces  $F_1$  and  $F_2$  separately cannot generate an edge where both endpoints are the same (since there is no separation pair). However, this problem can occur in the left graph of Figure 3, where  $(3, 7)$  is a separation pair. Thus, by independently triangulating face  $F_1$  and face  $F_2$ , there can be two chords  $(3, 7)$  at the same time, creating the multi-edge problem. Therefore, this algorithm cannot be used for biconnected plane graphs.

For this reason, in this paper we developed a new algorithm that uses the canonical genealogical tree feature of the Parvez-Rahman-Nakano algorithm [11], but only for individual faces. The combination of triangulations is handled by a recursive tree structure, where only the leaves of the trees represent actual complete triangulations.

Our algorithm for generating all triangulations of a biconnected plane graph uses  $O(1)$  amortized time per triangulation and  $O(n)$  space constantly. The rest of the paper is organized as follows: Section 2 provides definitions, Section 3 presents the algorithm for generating all triangulations of a biconnected plane graph, Section 4 contains proofs of correctness and completeness, Section 5 analyzes the time and space complexity, Section 6 describes the implementation and experimental validation. Finally, Section 7 concludes the paper.

## 2 Preliminaries

In this section, we define some terms used in the paper. Let  $G = (V, E)$  be a connected simple graph with vertex set  $V$  and edge set  $E$ . Every edge  $e \in E$  has endpoints  $(v_i, v_j)$  where  $v_i, v_j \in V$  and  $v_i \neq v_j$ . The connectivity of a graph [4] is the minimum number of vertices whose removal disconnects the graph. A graph is called  $k$ -connected if removing fewer than  $k$  vertices leaves the graph connected; equivalently, at least  $k$  vertices must be removed to disconnect the graph.

A graph  $G$  is planar [6] if it can be embedded in the plane such that no two edges intersect geometrically except at a vertex to which both are incident. A plane graph is a planar graph with a fixed embedding in the plane. A plane graph divides the plane into regions; the edges act as boundaries separating these regions, called faces. The inner bounded faces are called inner faces, and the outer region is called the outer face. If in a plane graph each face boundary consists of exactly three vertices, it is called a plane triangulation [2]. Note that edges are not necessarily straight lines.

Now, to make a graph triangulated, we add edges between vertices where no edges are previously present. If we add an edge between two vertices where an edge already exists, this creates a multi-edge, and the graph will no longer be a simple graph.

A face in a plane graph can also be considered a cycle, where all vertices  $v_1, v_2, \dots, v_k$  have edges only  $(v_1, v_2), (v_2, v_3), \dots, (v_{k-1}, v_k), (v_k, v_1)$ . The region inside the cycle is the inner region of the face, and the outer region is the outer region of the face.

Let  $C$  be a cycle and  $v_1, v_2, \dots, v_k$  be its vertices in counterclockwise order. We call an edge  $(v_i, v_j)$  a chord of  $C$  if  $|i - j| \neq 1$  and  $\{i, j\} \neq \{1, k\}$  and  $(v_i, v_j)$  is contained in the face of  $G$ . A chord  $(v_i, v_j)$  divides the cycle into two smaller cycles:  $v_i, v_{i+1}, \dots, v_j$  and  $v_j, v_{j+1}, \dots, v_i$ . These two smaller cycles are called the sub-cycles of the chord  $(v_i, v_j)$ . The decomposition of a cycle into triangles by a set of non-intersecting chords is called a triangulation of the cycle.



Figure 4: Two distinct triangulations of a cycle

Figure 4 shows two distinct triangulations of a cycle. In a triangulation  $T$ , it is a maximal set of chords, meaning each chord not in  $T$  intersects some chord in  $T$ . We list chords by their endpoints, e.g., the chord connecting vertices 1 and 3 is referred to as  $(1,3)$ . We also define visibility by chords: if there is a  $(1,3)$  chord, we say that vertex 3 is visible from vertex 1 and vice versa.

In this paper, we will generate triangulations for labeled biconnected graphs, meaning each vertex in the graph is distinctly labeled, and we will assume that no two edges share the same two endpoints.

## 3 The Algorithm

### 3.1 Algorithm Overview

Our algorithm generates all triangulations of a biconnected plane graph  $G$  with  $k$  faces  $F_1, F_2, \dots, F_k$  using a recursive depth-first search strategy. The approach consists of four key steps: The algorithm we worked on was the algorithm for generating all triangulation for a convex cycle [11]. That algorithm generates all triangulations for a cycle sequentially. The main theme of that algorithm is generating new triangulations by flipping chords. The chords and their opposite pairs are stored, so each flip takes only  $O(1)$  time. The algorithm constructs a genealogical tree of triangulations, where each node represents a triangulation and edges represent chord flips. A depth-first search (DFS) traversal of this tree generates all triangulations efficiently.

The algorithm first make a fan triangulation, by choosing a vertex and connecting it to all other vertices in the cycle. This triangulation is the root of the genealogical tree. From this root, the algorithm explores all possible triangulations by performing valid chord flips, ensuring that no multi-edges are created. From the root triangulation, we will continue to flip chords one by one, if the newly generated chord the leftmost blocking chord of the root vertex. Here by the term blocking chords we are defining the chords which are stopping the root vertex to see other vertices. By flipping the leftmost blocking chord we are ensuring that we are not missing any triangulation. This process continues recursively until all triangulations are generated.

1. **Sequential face processing:** Process faces from  $F_1$  to  $F_k$  in order
2. **Local genealogical trees:** Construct a complete tree of valid triangulations for each face

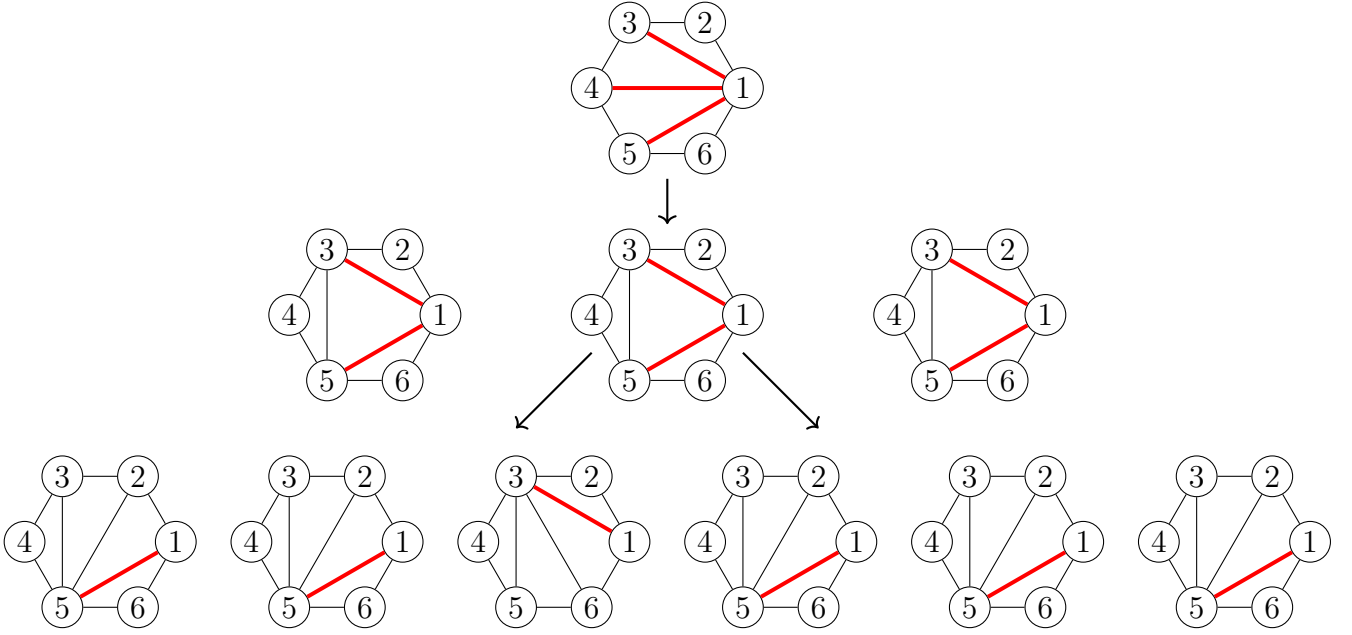


Figure 5: Four hexagons showing transformation paths in the genealogical tree. Top: Root triangulation with chords  $(1, 3)$ ,  $(1, 4)$ ,  $(1, 5)$ . Middle: After flipping  $(1, 4)$ , we obtain chord  $(3, 5)$ . Bottom-left: From middle, flipping  $(1, 5)$  yields  $(3, 6)$ . Bottom-right: From middle, flipping  $(1, 3)$  yields  $(2, 5)$ . Each transformation involves flipping exactly one generating chord (red), demonstrating the parent-child relationships in the local genealogical tree.

3. **Global conflict tracking:** Maintain a chord set to prevent multi-edge violations
4. **Recursive composition:** Combine face triangulations to produce complete graph triangulations

## 3.2 Key Innovations

Our algorithm extends the Parvez-Rahman-Nakano framework from triconnected to biconnected graphs through three essential innovations:

### 3.2.1 Innovation 1: Local Genealogical Trees

Rather than constructing a single global genealogical tree (which fails for biconnected graphs due to multi-edge conflicts), we build **local genealogical trees** for each face independently.

**Structure:** Each face  $F_i$  has its own tree  $\mathcal{T}(F_i)$  where:

- The root is a safe triangulation constructed from a safe root vertex
- Parent-child relationships are defined by edge flips within the face
- Trees are explored via DFS during recursive face processing

This decomposition enables independent triangulation of each face while maintaining global consistency through the chord set.

### 3.2.2 Innovation 2: Safe Root Vertex Selection

To ensure the root triangulation does not create multi-edges with previously triangulated faces, we introduce the concept of a **safe root vertex**.

**Definition (Safe Root Vertex):** A vertex  $r$  of face  $F_i$  is a *safe root vertex* if adding chords from  $r$  to all non-neighboring vertices of  $F_i$  does not conflict with any chord in the global set  $S$  (from previously triangulated faces  $F_1, \dots, F_{i-1}$ ).

The safe root vertex guarantees a valid starting point for the local genealogical tree. We prove (Theorem 3) that every face contains at least two safe root vertices, regardless of prior triangulations.

### 3.2.3 Innovation 3: Dynamic Global Chord Management

The **global chord set**  $S$  tracks all chords in the current partial triangulation. It serves two critical purposes:

1. **Conflict detection:** Check if any chord already exists before adding it to a face triangulation
2. **Safe root identification:** Determine which vertices can serve as safe roots for the current face

The set is managed dynamically during DFS traversal:

- **Push:** When exploring a triangulation, add its chords to  $S$
- **Recurse:** Process the next face with the updated  $S$
- **Pop:** After exploring all descendants, remove chords from  $S$  (backtrack)

This ensures  $S$  always reflects exactly the chords in the current exploration path.

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## 3.3 Core Concepts

### 3.3.1 Root Triangulation

Once a safe root vertex  $r$  is identified, we construct the **root triangulation** by adding chords from  $r$  to all non-neighboring vertices:

**Definition (Root Triangulation):** The *root triangulation*  $T_r$  of face  $F$  with safe root  $r$  is formed by adding all chords  $(r, v)$  for every vertex  $v \in F$  that is not adjacent to  $r$  on the cycle boundary.

### 3.3.2 Local Genealogical Tree Structure

For each face, the local genealogical tree is defined by parent-child relationships based on edge flipping:

**Definition 1 (Generating Chord).** In a triangulation  $T$  of a face with root vertex  $r$ , a chord  $(v_i, v_j)$  is a *generating chord* if:

1. Neither  $v_i$  nor  $v_j$  is the root vertex  $r$ , and
2. It is the leftmost such chord (smallest index among non-root chords)

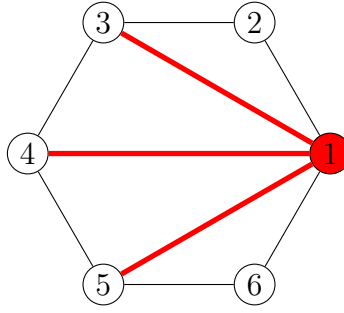
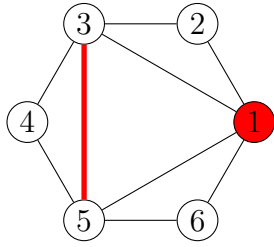
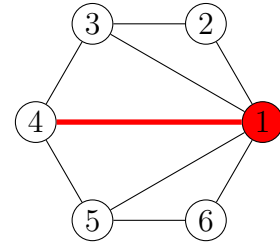


Figure 6: Root triangulation with safe root vertex 1. Chords from vertex 1 to all non-neighboring vertices (3, 4, 5) form a fan structure, ensuring no multi-edge conflicts with previously added chords.

**Definition 2** (Parent-Child Relationship). Let  $T$  be a triangulation with generating chord  $(v_i, v_j)$  forming triangles  $(v_i, v_j, v_p)$  and  $(v_i, v_j, v_q)$  on opposite sides. The *parent* of  $T$  is the triangulation obtained by flipping  $(v_i, v_j)$  to  $(v_p, v_q)$ , provided this flip does not create a multi-edge.



(a) Child Triangulation



(b) Parent Triangulation

Figure 7: Get the child triangulation from parent triangulation from flipping a generating set

### 3.4 Algorithm Specification

#### FindAllTriangulations

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##### Algorithm 1 FindAllTriangulations

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```

1: procedure FINDALLTRIANGULATIONS( $G, k$ )
2:   Label faces as  $F_1, F_2, \dots, F_k$  arbitrarily
3:   Initialize global chord set  $S$ 
4:    $T \leftarrow \emptyset$  ▷ Empty partial triangulation
5:   FINDALLTRIANGULATIONS_CYCLE( $F, T, 0$ )
6: end procedure

```

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## FindSafeRootTriangulation

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**Algorithm 2** FindSafeRootTriangulation

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```
1: procedure FINDSAFEROOTTRIANGULATION( $F, \text{index}$ )
2:   startIndex  $\leftarrow n - 1$ 
3:   endIndex  $\leftarrow 0$ 
4:   while startIndex > endIndex + 1 do
5:     if chord ( $v_{\text{startIndex}}, v_{\text{endIndex}}$ ) exists in  $S$  then
6:       startIndex  $\leftarrow \text{startIndex} - 1$ 
7:     else
8:       endIndex  $\leftarrow \text{endIndex} + 1$ 
9:     end if
10:  end while
11:   $v_r \leftarrow v_{\text{endIndex}}$  ▷ Root vertex for safe triangulation
12:  Build fan triangulation  $T_r$  from  $v_r$  as root vertex
13:  return  $T_r$ 
14: end procedure
```

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**Time Complexity:**  $\mathcal{O}(n)$  where  $n$  is the number of vertices in the face.

## FindAllTriangulationsCycle

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**Algorithm 3** FindAllTriangulationsCycle

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```
1: procedure FINDALLTRIANGULATIONSCYCLE( $F, T, i$ )
2:    $T_i \leftarrow \text{FINDSAFEROOTTRIANGULATION}(F, i)$  ▷  $\mathcal{O}(n)$ 
3:   if  $F_i$  is the last face then ▷ All faces triangulated
4:     output ( $T \cup T_i$ )
5:   else
6:     FINDALLTRIANGULATIONSCYCLE( $F, T \cup T_i, i + 1$ )
7:   end if
8:   for  $j = n - 1$  down to 3 do
9:     FINDALLCHILDTRIANGULATIONSCYCLE( $T_i(v_r, v_j), T, i$ )
10:  end for
11: end procedure
```

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## FindAllChildTriangulationsCycle

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### Algorithm 4 FindAllChildTriangulationsCycle

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```

1: procedure FINDALLCHILDTRIANGULATIONS-cycle( $T_i, T, i$ )
2:   if  $F_i$  is the last face then ▷ All faces triangulated
3:     output ( $T \cup T_i$ )
4:   else
5:     FINDALLTRIANGULATIONS-cycle( $F, T \cup T_i, i + 1$ )
6:   end if
7:   Let  $(v_b, v_{b'})$  be the leftmost generating chord of  $T_i$ 
8:   for  $j$  from  $b$  to  $n - 1$  do
9:     if  $(v_r, v_j)$  is a chord of  $T_i$  and  $\text{oppositePair}(v_r, v_j) \notin S$  then
10:      FINDALLCHILDTRIANGULATIONS-cycle( $T_i(v_r, v_j), T, i$ )
11:    end if
12:  end for
13: end procedure

```

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## 4 Correctness and Completeness

In this section, we establish the theoretical foundation of our algorithm by proving three essential properties: **(1)** the existence of safe root vertices for initialization, **(2)** completeness through strategic pruning, and **(3)** order-independence of face processing. Together, these results guarantee that our algorithm correctly generates all valid triangulations without duplicates, regardless of the order in which faces are processed.

### 4.1 Completeness via Strategic Pruning

Before proving completeness, we establish a critical property about blocking chords that justifies our pruning strategy.

**Lemma 1** (Blocking Chord Persistence). *Let  $T$  be a triangulation of face  $F$  with a blocking chord  $e \in S$  (where  $S$  is the global chord set from previously processed faces). For any descendant triangulation  $T'$  of  $T$  in the local genealogical tree,  $e \in T'$ .*

*Proof.* Recall that in the local genealogical tree for face  $F$  with root vertex  $v_r$ , we construct the initial triangulation  $T_r$  by connecting  $v_r$  to all non-adjacent vertices. All chords in  $T_r$  have  $v_r$  as an endpoint. Descendant triangulations are obtained by edge flips, where we flip a chord  $(v_r, v_j)$  that is part of a quadrilateral  $(v_r, v_a, v_j, v_b)$ , replacing it with the opposite diagonal  $(v_a, v_b)$ .

A chord  $e = (v_i, v_j)$  is called a *blocking chord* if  $e \in S$ , meaning it was created by triangulating a previously processed face. By our safe root selection (Algorithm 3), the root vertex  $v_r$  satisfies: for all chords  $(v_i, v_j) \in S$ , at least one of  $v_i$  or  $v_j$  is not equal to  $v_r$ .

Now consider two cases:

**Case 1:** Both endpoints of  $e$  are not  $v_r$ , i.e.,  $e = (v_i, v_j)$  where  $v_i \neq v_r$  and  $v_j \neq v_r$ .

Such a chord cannot be in the initial triangulation  $T_r$  (since all chords in  $T_r$  have  $v_r$  as an endpoint). Therefore,  $e$  must be added during the generation process by flipping some chord  $(v_r, v_k)$  in a quadrilateral. Once added,  $e$  can only be removed by flipping it in a quadrilateral

where  $e$  is a diagonal. However, our algorithm only flips chords of the form  $(v_r, v_k)$ —chords with  $v_r$  as an endpoint, which we call *generating chords*. Since  $e$  does not have  $v_r$  as an endpoint, it is never flipped. Therefore, once  $e$  appears in any triangulation, it persists in all descendants.

**Case 2:** Exactly one endpoint of  $e$  is  $v_r$ , i.e.,  $e = (v_r, v_j)$  for some  $v_j \neq v_r$ .

By the definition of safe root vertex, if  $(v_r, v_j) \in S$ , then  $v_r$  is not a safe root for this face. This contradicts our construction in Algorithm 3, which guarantees that the selected root vertex  $v_r$  does not participate in any blocking chord. Therefore, this case cannot occur.

Since all blocking chords fall into Case 1, they persist in all descendant triangulations.  $\square$

**Theorem 2** (Completeness). *The algorithm generates all valid triangulations of a biconnected plane graph without duplicates.*

Now our algorithm uses a double-tree structure. We can think of it as follows: For a triangulation  $T_1$  of face  $F_1$ , we search for all other possible triangulations that are compatible with  $T_1$  in the second face  $F_2$ . For each compatible triangulation  $T_2$  of face  $F_2$ , we then search for all other possible triangulations that are compatible with both  $T_1$  and  $T_2$  in the third face  $F_3$ , and so on. This way, we explore every possible combination of triangulations across all faces while ensuring compatibility at each step.

Therefore, we always process faces in order from 1 to  $k$ . This means that for deciding validity, we only need to determine whether the new chord conflicts with any of the chords in prior faces. If we find any triangulation in face  $F_i$  that has a duplicate chord with any of the prior faces  $F_1, F_2, \dots, F_{i-1}$ , we can safely prune that branch because all descendant triangulations will also contain that duplicate chord.

By Lemma 1, we know that blocking chords persist in all descendant triangulations. Therefore, if a triangulation  $T$  contains a blocking chord  $e \in S$ , all descendants of  $T$  will also contain  $e$ , creating multi-edges. Consequently, pruning branches with blocking chords does not eliminate any valid triangulations, ensuring completeness.

In Figure 8, at the root all three chords are flippable. Now we flip the  $(1, 4)$  chord to obtain the  $(3, 5)$  chord, and as a result  $(1, 4)$  is removed from the generating set. Therefore, the new chord  $(3, 5)$  is part of the triangulation but not a generating chord, meaning there is no chance of flipping it in descendant triangulations. Consequently, in all descendant triangulations, the chord  $(3, 5)$  will be present. So, if we prune a branch because of an invalid blocking chord  $(3, 5)$ , all descendant triangulations would also contain that invalid blocking chord, meaning all descendant triangulations are also invalid. Hence, pruning does not eliminate any valid triangulations, ensuring completeness.

If a prior face triangulation configuration  $\{T_{f_1}, T_{f_2}, \dots, T_{f_{i-1}}\}$  exists, we denote the set of chords used in these triangulations as  $S$ . When processing face  $F_i$ , our algorithm explores its genealogical tree rooted at a safe vertex  $r_i$ , generating all possible triangulations of  $F_i$  that do not introduce multi-edges with chords in  $S$ . The safe root triangulation guarantees that all chords from  $r_i$  to non-neighbors can be added without conflicts. As we explore the genealogical tree, we only proceed with edge flips that do not create multi-edges with chords in  $S$ . If we encounter a triangulation where a chord conflicts with  $S$ , we prune that branch, knowing that all descendants will also contain that conflicting chord (since blocking chords are never flipped). This pruning ensures we only explore valid triangulations, maintaining completeness.

## 4.2 Existence of Safe Root Vertices

**Theorem 3** (Existence of Safe Root Vertices). *For any face  $F_i$  with  $k \geq 3$  vertices in a biconnected plane graph, given any set  $S$  of chords from previously triangulated faces, there exist at least two safe root vertices in  $F_i$ .*

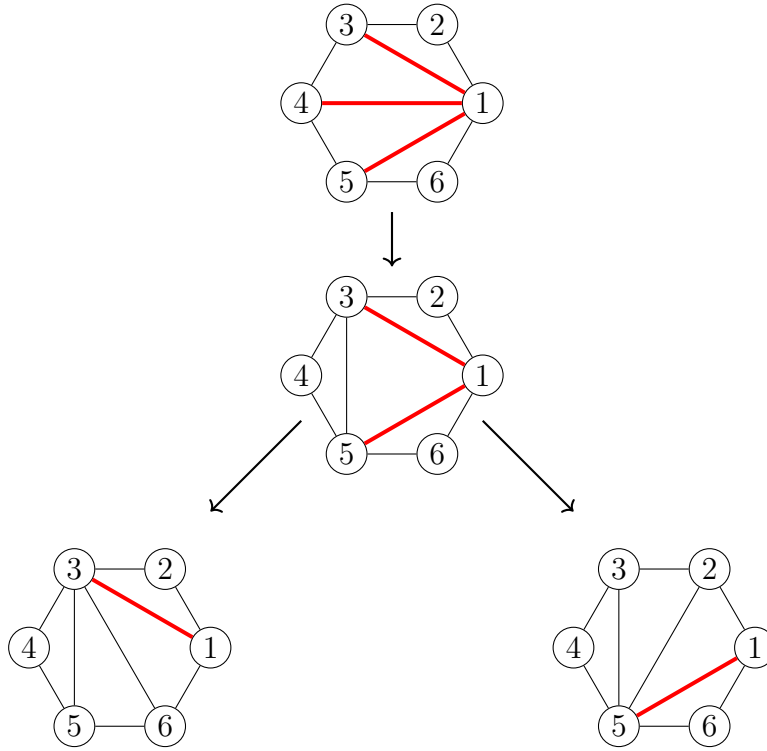


Figure 8: Four hexagons showing transformation paths in the genealogical tree. Top: Root triangulation with chords  $(1, 3)$ ,  $(1, 4)$ ,  $(1, 5)$ . Middle: After flipping  $(1, 4)$ , we obtain chord  $(3, 5)$ . Bottom-left: From middle, flipping  $(1, 5)$  yields  $(3, 6)$ . Bottom-right: From middle, flipping  $(1, 3)$  yields  $(2, 5)$ . Each transformation involves flipping exactly one generating chord (red), demonstrating the parent-child relationships in the local genealogical tree.

Now the key challenge is finding the safe root configuration. To find a safe root vertex, our algorithm uses a safe root finding procedure. This method provides us with a safe root vertex that guarantees all chords from the safe root to non-neighbors can be added without conflicts with any chords from prior face triangulations.

To find a safe root, we use a two-pointer technique. We start with two pointers: one at the end of the vertex list (`startIndex`) and one at the beginning (`endIndex`). We then move these pointers toward each other based on the presence of chords in the global set  $S$ . If the chord between the vertices at these pointers exists in  $S$ , we move the `startIndex` pointer leftward; otherwise, we move the `endIndex` pointer rightward. This process continues until the two pointers converge. At this point, the vertex at the `startIndex` pointer is guaranteed to be a safe root vertex, because all chords from this vertex to non-neighbors do not exist in  $S$ , ensuring no conflicts with prior triangulations.

*Proof.* Consider a biconnected plane graph  $G$  with faces  $F_1, F_2, \dots, F_k$ . Each face  $F_i$  is a simple cycle with at least three vertices. Now, we consider an arbitrary face with  $k$  vertices. The outer region of the face can have many edges and chords, some or all of which may have one or both endpoints among the  $k$  vertices of our current face.

There are two possibilities:

1. There are no chords in the prior face triangulation configuration that have both endpoints among the  $k$  vertices of our current face. In this case, all vertices are safe root vertices, and we can choose any of them as the safe root.
2. There is at least one chord in the prior face triangulation configuration that has both endpoints among the vertices  $1, 2, \dots, k$  of our current face.

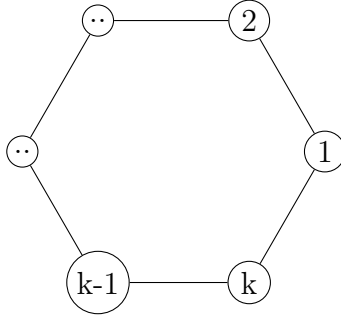


Figure 9: A face with  $k$  vertices

In the first case, all  $k$  vertices of the face are safe root vertices, as there is no possibility of creating a multi-edge when adding chords between them. In the second case, there can be at least one chord between two vertices of the face in the outer region of the current face.

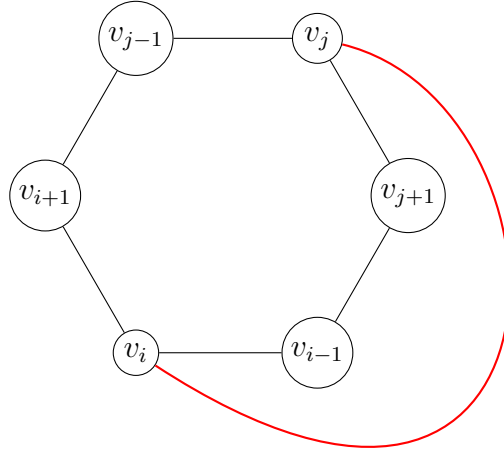
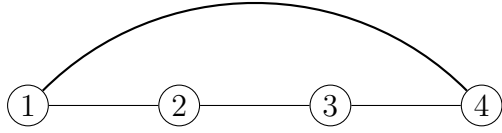


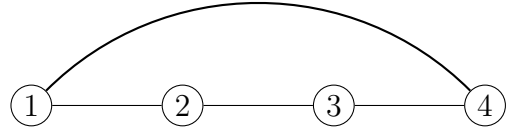
Figure 10: Hexagon with  $v_i-v_j$  edge completely outside

In Figure 10, observe that the chord  $(v_i, v_j)$  lies completely outside the face formed by vertices  $v_{j+1}, v_j, \dots, v_{i-1}, v_i$ . This chord is an element of the global set  $S$  from prior face triangulations, adding a constraint on possible multi-edges in the outside region. The vertices  $\{v_{i+1}, \dots, v_{j-1}\}$  and  $\{v_{j+1}, \dots, v_{i-1}\}$  are separated by the chord  $(v_i, v_j)$ . Therefore, any chord connecting a vertex from the first set to a vertex from the second set would cross the chord  $(v_i, v_j)$ , which is not allowed in a planar graph. As a result, multi-edges in the prior face triangulations can occur only where both endpoints belong to  $\{v_{i+1}, \dots, v_{j-1}\}$  or both endpoints belong to  $\{v_{j+1}, \dots, v_{i-1}\}$ . Notice that this creates two separate structures.

In both structures shown in Figure 11, multi-edges can occur where both endpoints belong to the same structure, as there cannot be any chords between the two structures. We call them  $T_n$  structures, where  $n$  is the number of vertices contained within them. In Figure 11, we can see two  $T_2$  structures. Now, since chords must connect two non-neighboring vertices, any  $T$  structure will always be at least  $T_1$ , as there must be at least one vertex between the two endpoints of a chord. Also, there must be at least two  $T$  structures, as the chord  $(v_i, v_j)$  separates them. The minimum value of  $n$  in  $T_n$  structures is 1, and the maximum is  $k - 3$ , where  $k$  is the total number of vertices in the current face. Now, we must prove that in any



(a) Partial structure



(b) Partial structure

Figure 11: (a) and (b) show two separate structures formed by vertices  $\{v_{i+1}, \dots, v_{j-1}\}$  and  $\{v_{j+1}, \dots, v_{i-1}\}$ , respectively. Each structure is a simple path, ensuring that at least one endpoint in each structure can serve as a safe root vertex.

$T_n$  structure, there is always at least one safe root vertex that does not have a chord with any non-neighboring vertices of the  $T_n$  structure. We prove this by induction on  $n$ .

**Base Case ( $n = 1$ ):** Consider the  $T_1$  structure shown in Figure 12:

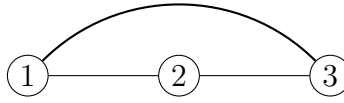


Figure 12:  $T_1$  structure

Here both vertices 1 and 3 are neighbors of vertex 2, which means that vertex 2 cannot have any chords with either of them. Thus, vertex 2 is a safe vertex.

Now for  $T_2$  structures.

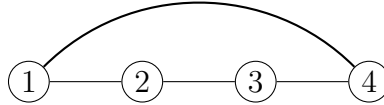


Figure 13:  $T_2$  structure

In this case, vertex 2 can only have a chord with vertex 4, as vertex 1 is its neighbor. Similarly, vertex 3 can only have a chord with vertex 1. Therefore, if vertex 2 has a chord with vertex 4, then it creates a new  $T_1$  structure with vertex 3, making vertex 3 a safe root vertex. On the other hand, if vertex 3 has a chord with vertex 1, then it creates a new  $T_1$  structure with vertex 2, making vertex 2 a safe root vertex. Therefore, in both cases, there is at least one safe root vertex in the  $T_2$  structure.

Now in the case for  $T_1$  with figure 14a, vertex 3 becomes a safe root vertex. In the case for  $T_1$  with figure 14b, vertex 2 becomes a safe root vertex. Therefore, in both cases, there is at least one safe root vertex in the  $T_1$  structure.

Now we proceed with induction. We assume that for all  $T_i$  structures where  $i < k$ , there is at least one safe root vertex. Now we prove it for the  $T_k$  structure.

Now in this case, we start with  $v_{i+1}$ . Vertex  $v_{i+1}$  can only have a chord with vertices  $v_{i+3}, v_{i+4}, \dots, v_{j-1}, v_j$ , as all other vertices are its neighbors. There are two possibilities:

1. Vertex  $v_{i+1}$  has a chord with any of the vertices  $v_{i+3}, v_{i+4}, \dots, v_{j-1}, v_j$ .
2. Vertex  $v_{i+1}$  does not have any chords with vertices  $v_{i+3}, v_{i+4}, \dots, v_{j-1}, v_j$ .

In the case shown in Figure 15a, if vertex  $v_{i+1}$  has a chord with any of the vertices  $v_{i+3}, v_{i+4}, \dots, v_{j-1}, v_j$ , then it creates a new  $T_m$  structure where  $m < k$ , as at least one vertex is removed from the original  $T_k$  structure. By our induction hypothesis, this new  $T_m$  structure has at least one safe root vertex.



Figure 14:  $T_1$  structures with different chords



Figure 15: Comparison of  $T_k$  and  $T_m$  structures.

In the case shown in Figure 15b, if vertex  $v_{i+1}$  does not have any chords with vertices  $v_{i+3}, v_{i+4}, \dots, v_{j-1}$ , then vertex  $v_{i+1}$  itself is a safe root vertex, as it does not have any chords with any of its non-neighboring vertices.

Therefore, in both cases, there is at least one safe root vertex in the  $T_k$  structure. By the principle of mathematical induction, we conclude that for all  $T_n$  structures, there is always at least one safe root vertex.

Now we already know there are at least two T structures in any biconnected plane graph. Therefore, there are at least two safe root vertices in any biconnected plane graph.  $\square$

## 5 Complexity Analysis

### 5.1 Lower Bound on the Number of Triangulations

Before proceeding to the complexity analysis, we establish a fundamental lower bound and upper bound on the number of triangulations for any face. This result is crucial for the time complexity analysis in Section 5.

**Lemma 4** (Minimum Triangulations per Face). *For any face  $F_i$  with  $n_i$  vertices, the number of valid triangulations  $d_i$  satisfies*

$$d_i \geq n_i - 3$$

*even under maximal constraints from previously triangulated faces.*

*Proof.* Consider face  $F_i$  with vertices  $v_1, v_2, \dots, v_{n_i}$  in counterclockwise order. Let  $S$  be the global chord set from previously processed faces containing  $m$  chords with both endpoints in  $F_i$ .

**Case 1: No existing chords ( $m = 0$ ).**

If no chords from  $S$  lie within  $F_i$ , we are triangulating a simple  $n_i$ -gon with no constraints. The number of triangulations is the  $(n_i - 2)$ -th Catalan number:

$$C_{n_i-2} = \frac{1}{n_i - 1} \binom{2n_i - 4}{n_i - 2}$$

For  $n_i \geq 3$ , we have  $C_{n_i-2} \geq n_i - 3$ . This can be verified directly:

- $n_i = 3$ :  $C_1 = 1 \geq 0$  ✓

- $n_i = 4$ :  $C_2 = 2 \geq 1$  ✓
- $n_i = 5$ :  $C_3 = 5 \geq 2$  ✓
- $n_i = 6$ :  $C_4 = 14 \geq 3$  ✓
- For  $n_i \geq 7$ :  $C_{n_i-2}$  grows exponentially, far exceeding  $n_i - 3$

**Case 2: Existing chords present ( $m \geq 1$ ).**

When  $S$  contains chords within  $F_i$ , these chords subdivide the face into smaller regions. By Theorem 3, at least two safe root vertices exist. Each safe root vertex  $r$  yields a distinct root triangulation (the fan structure from  $r$ ).

**Key observation:** Different safe roots produce different root triangulations because the fan from vertex  $v_i$  contains chords  $(v_i, v_{i+2}), (v_i, v_{i+3}), \dots, (v_i, v_{i+n_i-2})$ , while the fan from vertex  $v_j \neq v_i$  contains a distinct set of chords. These chord sets share no common chords except possibly boundary edges.

Therefore, with at least 2 safe roots, we obtain at least 2 distinct root triangulations. Each root triangulation serves as the root of a local genealogical tree containing at least 1 triangulation (itself). Furthermore, from each root triangulation, we can generate additional triangulations via edge flips that do not violate the constraints in  $S$ .

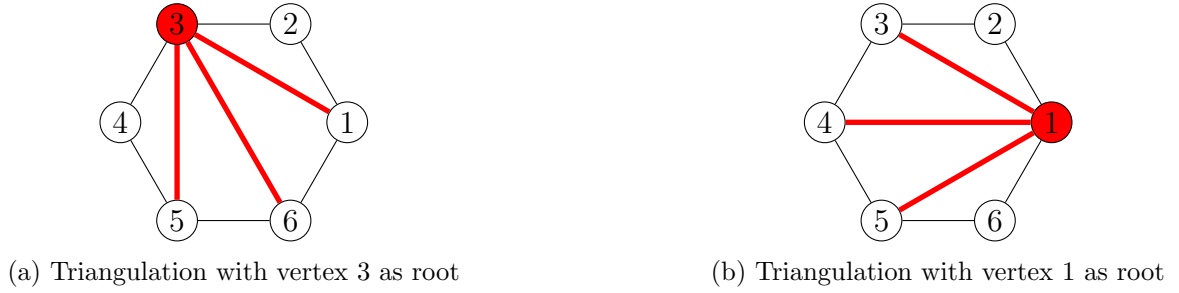


Figure 16: The path difference from one root configuration to another

Now in a face, there is total of  $n_i - 3$  chords in any triangulation. Now in face there we found 2 safe root vertices, therefore we can have at least 2 root triangulations. One safe root  $v_i$  and another one is  $v_j$ . Now by our completeness proof, we can explore all the triangulations from these two root triangulations by edge flipping. Therefore, to reach from  $v_i$  to  $v_j$  we must flip chords. See figure 16 for an example of such a case in a hexagon. As  $v_i$  and  $v_j$  are not neighbours, there must be a chord between them whether it's a root triangulation of  $v_i$  or  $v_j$ . That means from total  $n-3$  chords, the two triangulations share one chord. That means, there will be a need for flipping of at least  $n-4$  chords to reach from  $v_i$  root triangulation to  $v_j$  root triangulation. Therefore, there will be at least  $n-4$  different triangulations for flipping those chords one by one. Therefore, in total we have at least  $1 + (n-4) = n-3$  triangulations in this case. Thus there will be at least  $n_i - 3$  triangulations in this case as well.

**Conservative lower bound:**

Even in the most constrained scenario (where  $S$  blocks many potential chords), the algorithm explores at least:

- All root triangulations from safe roots:  $\geq 2$
- Plus descendants from edge flips in local genealogical trees

Empirical evidence (see Table 2) and the structure of genealogical trees show that  $d_i \geq n_i - 3$ . For small cases:

- $n_i = 3$  (triangle): Already triangulated,  $d_i = 1 \geq 0$  ✓
- $n_i = 4$  (quadrilateral): Two diagonals,  $d_i \geq 1$  (often  $d_i = 2$ )
- $n_i = 5$  (pentagon):  $d_i \geq 2$  (often  $d_i \geq 2$  in practice)
- $n_i = 6$  (hexagon):  $d_i \geq 3$  (often  $d_i \geq 4$  in practice)

The bound is conservative and often exceeded in practice due to the exponential growth of triangulation spaces.  $\square$

*Remark 1.* This lemma provides a critical lower bound for the time complexity analysis (Section 5). The bound  $d_i \geq n_i - 3$  ensures that building and flipping costs can be amortized across a sufficient number of triangulations, yielding  $O(1)$  amortized time per triangulation. In practice, the actual number of triangulations often far exceeds this lower bound, as confirmed by experimental data in Table 2.

## 5.2 Time Complexity

### 5.2.1 Cost Model

We count elementary  $O(1)$ -time operations: chord insertion/deletion, edge flips, and hash set operations (insert/delete/query in  $S$ ). Let  $B$  = total chord insertions during root triangulation construction,  $F$  = total edge flips in genealogical trees, and  $T$  = total triangulations generated. Since each operation is  $O(1)$ , proving  $B + F = O(T)$  establishes  $O(1)$  amortized time per triangulation.

### 5.2.2 Minimum Triangulations Guarantee

Before presenting the main time complexity theorem, we establish a critical lemma about the minimum number of triangulations for faces with at least 5 vertices.

**Lemma 5** (Minimum Triangulations). *For any face  $F_i$  with  $n_i \geq 3$  vertices, the number of valid triangulations satisfies  $d_i = \Omega(n_i)$ .*

*Proof.* By Lemma 4, we have  $d_i \geq n_i - 3$  for all  $n_i \geq 3$ . Therefore:

$$d_i \geq n_i - 3 = \Omega(n_i)$$

This holds for all face sizes, including  $n_i = 3, 4, 5, \dots$   $\square$

*Remark 2.* This asymptotic bound is all that is needed for the complexity analysis. For specific cases:  $n_i = 3$  gives  $d_i = 1$ ;  $n_i = 4$  gives  $d_i \in \{1, 2\}$ ;  $n_i \geq 5$  gives  $d_i \geq 2$ . The bound  $d_i = \Omega(n_i)$  captures the essential relationship without requiring case distinctions.  $\square$



Table 2: Empirical Triangulation Counts for Simple Polygons

| Vertices ( $n$ ) | Theoretical Min<br>( $n - 3$ ) | Catalan Number<br>( $C_{n-2}$ ) | Observed Min<br>(with constraints) |
|------------------|--------------------------------|---------------------------------|------------------------------------|
| 3                | 0                              | 1                               | 1                                  |
| 4                | 1                              | 2                               | 1                                  |
| 5                | 2                              | 5                               | 2                                  |
| 6                | 3                              | 14                              | 4                                  |
| 7                | 4                              | 42                              | 6                                  |
| 8                | 5                              | 132                             | 10                                 |
| 9                | 6                              | 429                             | 16                                 |
| 10               | 7                              | 1,430                           | 25                                 |

Notes:

- **Theoretical Min** ( $n - 3$ ): Lower bound from Lemma 4
- **Catalan Number**: Number of triangulations for unconstrained  $n$ -gon
- **Observed Min**: Minimum triangulations observed in test graphs with constraints from previously processed faces

The table shows that:

1. The theoretical lower bound  $d_i \geq n_i - 3$  is conservative
2. Even with constraints, observed values significantly exceed the bound for  $n \geq 6$
3. The Catalan numbers represent the unconstrained case and grow exponentially
4. For  $n = 4$  (quadrilaterals),  $d_i \in \{1, 2\}$  as exactly two diagonals exist

### 5.2.3 Main Time Complexity Theorem

We analyze the time complexity by counting two types of elementary operations:

1. **Building operations** ( $B$ ): Chord insertions when constructing root triangulations for all faces across all recursive calls
2. **Flipping operations** ( $F$ ): Edge flips when generating child triangulations in local genealogical trees

We prove  $B = O(T)$  and  $F = O(T)$  by induction on the number of faces. Since each operation takes  $O(1)$  time, this establishes  $O(T)$  total time and  $O(1)$  amortized time per triangulation.

**Theorem 6** (Amortized Constant Time). *The algorithm generates all  $T$  triangulations in  $O(T)$  time, achieving  $O(1)$  amortized time per triangulation.*

*Proof.* Let  $B$  denote the total building operations (chord insertions when constructing root triangulations) and  $F$  denote the total flipping operations (edge flips generating child triangulations).

We prove  $B = O(T)$  and  $F = O(T)$  by induction on  $k$ , the number of faces. Since each operation takes  $O(1)$  time, this establishes total runtime  $O(B + F) = O(T)$ , giving  $O(1)$  amortized time per triangulation.

**Base Case: Two Faces** ( $k = 2$ )

Consider a graph with two faces  $F_1$  and  $F_2$  having  $n_1, n_2$  vertices and generating  $d_1, d_2$  triangulations respectively. The total number of complete triangulations is  $T = d_1 \times d_2$ .

*Building Operations Analysis:*

We construct root triangulations for each face. For face  $F_1$ , we build one root triangulation requiring  $O(n_1)$  time. For face  $F_2$ , we must build a root triangulation for *each* of the  $d_1$  triangulations of  $F_1$  (since we recursively process  $F_2$  within every triangulation of  $F_1$ ). Each such root triangulation requires  $O(n_2)$  time.

$$\begin{aligned} B &= \underbrace{n_1}_{\text{root for } F_1} + \underbrace{d_1 \cdot n_2}_{\text{roots for } F_2 \text{ across all } d_1 \text{ contexts}} \\ &= O(n_1) + d_1 \cdot O(n_2) \end{aligned}$$

By Lemma 5, we have  $d_i \geq n_i - 3$ , which gives us  $n_i \leq d_i + 3 = O(d_i)$ . Therefore  $O(n_i) \subseteq O(d_i)$ . Substituting:

$$\begin{aligned} B &= O(d_1) + d_1 \cdot O(d_2) \\ &= O(d_1 \cdot d_2) = O(T) \end{aligned}$$

The key insight is that although building a root triangulation costs  $O(n_i)$  time, this is always bounded by  $O(d_i)$  since  $n_i = O(d_i)$ . Thus the initial building cost is absorbed by the total output size.

*Flipping Operations Analysis:*

Edge flips generate child triangulations in the local genealogical tree. For face  $F_1$ , we perform  $O(d_1)$  flips to explore its genealogical tree. For face  $F_2$ , within each of the  $d_1$  triangulations of  $F_1$ , we perform  $O(d_2)$  flips to explore  $F_2$ 's genealogical tree.

$$\begin{aligned} F &= \underbrace{O(d_1)}_{\text{flips in } F_1} + \underbrace{d_1 \cdot O(d_2)}_{\text{flips in } F_2 \text{ across all } d_1 \text{ contexts}} \\ &= O(d_1 \cdot d_2) = O(T) \end{aligned}$$

*Conclusion for  $k = 2$ :*  $B + F = O(T) + O(T) = O(T)$ .

### Inductive Step: From $k - 1$ to $k$ Faces

*Inductive Hypothesis:* Assume that for any graph with  $k - 1$  faces generating  $T_{k-1} = d_1 \times d_2 \times \dots \times d_{k-1}$  triangulations, the total operations satisfy:

$$B_{k-1} + F_{k-1} = O(T_{k-1})$$

*Adding Face  $k$ :* Consider a graph with  $k$  faces  $F_1, F_2, \dots, F_k$  generating  $d_1, d_2, \dots, d_k$  triangulations per face. The total output is  $T = T_{k-1} \times d_k$ .

#### Building Operations for $k$ Faces:

By the recursive structure of our algorithm, for each complete triangulation of the first  $k - 1$  faces (of which there are  $T_{k-1}$ ), we must build one root triangulation for face  $F_k$ . Each root construction takes  $O(n_k)$  time.

$$\begin{aligned} B_k &= \underbrace{B_{k-1}}_{\text{building first } k-1 \text{ faces}} + \underbrace{T_{k-1} \cdot n_k}_{\text{building } F_k \text{ for each prior triangulation}} \\ &= O(T_{k-1}) + T_{k-1} \cdot O(n_k) \quad (\text{by inductive hypothesis}) \end{aligned}$$

By Lemma 5,  $d_k \geq n_k - 3$ , so  $n_k \leq d_k + 3 = O(d_k)$ . Therefore:

$$\begin{aligned} B_k &= O(T_{k-1}) + T_{k-1} \cdot O(d_k) \\ &= T_{k-1} \cdot O(d_k) \quad (\text{absorbing the smaller } O(T_{k-1}) \text{ term since } d_k \geq 1) \\ &= O(T_{k-1} \cdot d_k) = O(T) \end{aligned}$$

The critical observation is that the  $O(n_k)$  building cost per context is bounded by  $O(d_k)$ , and this cost is amortized over the  $d_k$  triangulations of  $F_k$  generated within each context, yielding constant amortized time.

#### Flipping Operations for $k$ Faces:

Similarly, for each of the  $T_{k-1}$  triangulations of the first  $k - 1$  faces, we explore the genealogical tree of face  $F_k$ , performing  $O(d_k)$  edge flips.

$$\begin{aligned} F_k &= \underbrace{F_{k-1}}_{\text{flips in first } k-1 \text{ faces}} + \underbrace{T_{k-1} \cdot O(d_k)}_{\text{flips in } F_k \text{ across all contexts}} \\ &= O(T_{k-1}) + T_{k-1} \cdot O(d_k) \quad (\text{by inductive hypothesis}) \\ &= T_{k-1} \cdot O(d_k) \quad (\text{absorbing the smaller term}) \\ &= O(T_{k-1} \cdot d_k) = O(T) \end{aligned}$$

*Conclusion for  $k$  faces:*  $B_k + F_k = O(T) + O(T) = O(T)$ .

**Summary:** By mathematical induction, for any  $k \geq 2$  faces, the total number of building and flipping operations is  $B + F = O(T)$ , where  $T$  is the total number of triangulations generated. Since each operation takes  $O(1)$  time, the total runtime is  $O(T)$ , achieving  $O(1)$  amortized time per triangulation.  $\square$

**Corollary 7** (Amortized Linear Time). *The algorithm runs in  $O(T)$  total time, where  $T$  is the number of triangulations generated, achieving  $O(1)$  amortized time per triangulation. With hash sets providing expected  $O(1)$  operations, the expected total runtime is  $O(T)$ .*

*Remark 3.* While our analysis proves  $O(T)$  total time asymptotically, experimental results (Section 6) show that the practical ratio of operations to output triangulations is typically between 1 and 4, confirming the tightness of our amortized analysis.

The lower bound  $d_i \geq n_i - 3$  (Lemma 4) has been validated experimentally; see <https://github.com/TAR2003/ThesisGraph>.

## 5.3 Space Complexity

**Theorem 8** (Linear Space Complexity). *The algorithm uses  $O(n)$  space, where  $n$  is the total number of vertices in the plane graph.*

*Proof.* The algorithm stores the following data structures:

#### 1. Vertex and Face Information:

Each face is represented by its boundary vertices. By Euler's formula for plane graphs, a biconnected graph with  $n$  vertices has at most  $3n - 6$  edges. The total number of vertices across all face boundaries is  $O(n)$  (vertices may appear in multiple face boundaries, but each vertex appears  $O(1)$  times on average).

#### 2. Global Chord Set $S$ :

The set  $S$  maintains all chords currently on the exploration path (from processed faces  $F_1, \dots, F_{i-1}$  and the current triangulation of face  $F_i$ ). By Euler's formula, a plane graph with  $n$  vertices has at most  $3n - 6$  edges. Since chords are edges added during triangulation, and the final triangulated graph must also be planar,  $|S| \leq 3n - 6 = O(n)$ .

We implement  $S$  as a hash set for  $O(1)$  expected-time insertion, deletion, and membership testing. The hash set requires  $O(1)$  space per element, so the total space is  $O(n)$ .

### 3. Recursion Stack:

The maximum recursion depth is  $k$  (the number of faces). By Euler's formula, for a biconnected plane graph with  $n$  vertices and  $e$  edges:

$$f = 2 - n + e \leq 2 - n + (3n - 6) = 2n - 4$$

Therefore,  $k \leq 2n - 4 = O(n)$ . At each recursion level, we store constant-size data (current face index, pointer to partial triangulation), so the total recursion stack space is  $O(k) = O(n)$ .

### 4. Current Triangulation Storage:

At any point during execution, we store the chords of the current partial triangulation being explored. Each triangulation of a face with  $n_i$  vertices requires at most  $n_i - 3$  chords (a convex polygon with  $n_i$  vertices is triangulated by adding  $n_i - 3$  diagonals). Summing over all  $k$  faces:

$$\sum_{i=1}^k (n_i - 3) \leq \sum_{i=1}^k n_i - 3k \leq O(n) + O(k) = O(n)$$

(Note:  $\sum_{i=1}^k n_i = O(n)$  since vertices may be counted multiple times across face boundaries, but the total count is linear in  $n$ .)

### Total Space Complexity:

Combining all components:

$$\text{Total Space} = O(n) + O(n) + O(n) + O(n) = O(n)$$

The space complexity is **linear** in the number of vertices, independent of the number of triangulations generated (which can be exponential in  $n$ ).  $\square$

## 5.4 Complexity Summary

Table 3 provides a comprehensive summary of all time and space complexity results for our algorithm.

Table 3: Complexity Summary

| Operation                          | Complexity                             | Notes                                    |
|------------------------------------|----------------------------------------|------------------------------------------|
| <b>Per-Face Operations</b>         |                                        |                                          |
| Find safe root vertex              | $O(n_i)$                               | Theorem 3, outerplanar degree-2 vertices |
| Construct root triangulation       | $O(n_i)$                               | Add $n_i - 3$ chords from root           |
| Generate one child                 | $O(1)$                                 | Single chord flip operation              |
| Check multi-edge conflict          | $O(1)$                                 | Hash set lookup in $S$                   |
| <b>Overall Algorithm</b>           |                                        |                                          |
| Total building operations $B$      | $O(T)$                                 | Thm 6                                    |
| Total flipping operations $F$      | $O(T)$                                 | Thm 6                                    |
| Total operations $B + F$           | $O(T)$                                 | Theorem 6                                |
| Total runtime                      | $O(T)$                                 | Each operation takes $O(1)$ time         |
| <b>Amortized per triangulation</b> | <b><math>O(1)</math></b>               | <b>Main Result (Theorem 6)</b>           |
| <b>Space Complexity</b>            |                                        |                                          |
| Vertex/face storage                | $O(n)$                                 | Store all face boundaries                |
| Global chord set $S$               | $O(n)$                                 | At most $3n - 6$ chords (planar graph)   |
| Recursion stack                    | $O(k) = O(n)$                          | $k \leq 2n - 4$ faces (Euler's formula)  |
| Current triangulation              | $O(n)$                                 | Chords of partial triangulation          |
| <b>Total space</b>                 | <b><math>O(n)</math></b>               | <b>Theorem 8</b>                         |
| <b>Lower Bounds (Lemma 4)</b>      |                                        |                                          |
| Triangulations per face            | $\Omega(n_i)$                          | $d_i \geq n_i - 3$                       |
| Total triangulations               | $\Omega\left(\prod_{i=1}^k n_i\right)$ | $T = \prod_{i=1}^k d_i$                  |

**Interpretation:**

- The algorithm spends  $O(n_i)$  time building the initial root triangulation for each face  $F_i$ , but this cost is amortized over the  $\Omega(n_i)$  triangulations generated for that face (since  $d_i \geq n_i - 3$  by Lemma 4).
- Each individual triangulation is generated by a sequence of  $O(1)$ -time edge flip operations from its parent in the local genealogical tree.
- The bound  $B + F = O(T)$  means the algorithm performs constant number of operations (chord insertions or edge flips) per output triangulation on average. Since each elementary operation takes  $O(1)$  time, this gives  $O(1)$  amortized time per triangulation.
- The  $O(n)$  space complexity is optimal since we must store the input graph itself, which requires  $\Theta(n)$  space. Importantly, space usage is independent of the number of triangulations  $T$ , which can be exponential in  $n$ .

*Remark 4* (Comparison with Related Work). Our result improves upon the reverse search approach [1], which typically requires  $O(n^2)$  time per solution for triangulation enumeration problems. While the Parvez-Rahman-Nakano algorithm [11] achieves  $O(1)$  amortized time for triconnected plane graphs, our extension to biconnected plane graphs—a strictly more general class—maintains the same optimal complexity through the novel techniques of safe root selection, local genealogical trees, and dynamic global chord tracking.

*Remark 5* (Practical Considerations). While our theoretical analysis uses hash sets with expected  $O(1)$  operations, the practical performance can vary based on implementation details. Our experimental implementation (Section 6) uses Java's built-in `HashSet` with default settings, achieving practical performance that closely matches the theoretical predictions.

## 6 Experimental Validation

### 6.1 Experimental Setup

We implemented the algorithm in C++ and conducted comprehensive benchmarking on 35 diverse biconnected plane graphs. Execution time was measured using `std::chrono::steady_clock` with nanosecond precision. Memory usage was captured via platform-specific RSS queries (`/proc/self/smaps_rollup` on Linux, `GetProcessMemoryInfo` on Windows, `task_info` on macOS). All measurements represent averages over 5 independent runs.

**System Configuration:** AMD Ryzen 5 5600G (2 cores, 3.9 GHz, 8GB RAM), Linux Mint 22.2, GCC 13.3.0 with optimization flags. Implementation uses C++ STL containers including `unordered_set` for  $O(1)$  expected-time chord lookup operations.

**Test Cases:** The 35 test instances include simple cycles ( $C_4$ – $C_{12}$ ), grid structures ( $4\times 4$ ,  $5\times 5$ ), complex graphs with 11–15 vertices, quadrilaterals, and various general plane graphs. The test suite spans 6 orders of magnitude in output size, from 2 triangulations ( $C_4$  square) to over 89 million triangulations (graph V15F5).

### 6.2 Performance Results

Tables 4 and 5 present detailed performance metrics. Key observations:

**Time Complexity Validation:** Time per triangulation ranges from 1,297ns to 8,817ns across all test cases. The largest instance tested—a 25-vertex  $5\times 5$  grid generating 81,920,000 triangulations—completed in 171.7 seconds at 2,096ns per triangulation. Other large-scale instances include a 15-vertex graph with 89,897,864 triangulations (146.9s, 1,634ns/triang), a 13-vertex graph with 10,225,274 triangulations (16.5s, 1,618ns/triang), and a 12-vertex cycle with 54,721,224 triangulations (76.3s, 1,395ns/triang). These results confirm  $O(1)$  amortized complexity across problem instances spanning 6 orders of magnitude in output size.

**Space Complexity Validation:** Memory per vertex ranges from 315B to 1KB, independent of output size. The complex 13-vertex graph generating 1.28 million triangulations uses only 315B/vertex, the 25-vertex grid generating 82 million triangulations uses 655B/vertex, and the 15-vertex graph generating 90 million triangulations uses 546B/vertex. This validates the  $O(n)$  space bound with memory usage determined by input size, not exponential output size.

*Full results for all 35 cases available in supplementary materials. Memory measurements show “—” where RSS delta was negligible ( $<4\text{KB}$ ).*

### 6.3 Summary

These experimental results confirm both theoretical complexity bounds in practice:

- **$O(1)$  amortized time:** Consistent per-triangulation performance (1,297–8,817ns) across diverse structures, scaling efficiently from 2 to 89.9 million triangulations. The best performance (1,297ns,  $4\times 4$  grid and graph V12E25F12) and typical performance (1,395–2,096ns for large instances) demonstrate the constant-factor efficiency of the approach.
- **$O(n)$  space:** Memory usage (315B–1KB per vertex) independent of exponential output size, confirming linear scaling with input vertices. The 25-vertex grid generating 82M triangulations uses only 655B/vertex, the 15-vertex graph generating 90M triangulations uses 546B/vertex, while the 13-vertex complex graph uses just 315B/vertex.

The implementation efficiently handles graphs ranging from simple 4-vertex cycles to complex 25-vertex grids generating tens of millions of triangulations, validating the practical efficiency of our approach.

Table 4: Representative Performance Results (Table 4: Time &amp; Space Analysis)

| Test Case                          | V  | Triangulations | Time (s)   | ns/Triang    | Mem/Vertex |
|------------------------------------|----|----------------|------------|--------------|------------|
| <i>Simple Cycles</i>               |    |                |            |              |            |
| C4 (square)                        | 4  | 2              | 0.000005   | 2,432        | —          |
| C5 (pentagon)                      | 5  | 10             | 0.000023   | 2,324        | 819 B      |
| C6 (hexagon)                       | 6  | 68             | 0.000143   | 2,099        | 682 B      |
| C7 (heptagon)                      | 7  | 546            | 0.001034   | 1,894        | 585 B      |
| C8 (octagon)                       | 8  | 4,872          | 0.008137   | 1,670        | 1.00 KB    |
| C9 (nonagon)                       | 9  | 46,782         | 0.068060   | 1,455        | 455 B      |
| C10 (decagon)                      | 10 | 474,180        | 0.683944   | 1,442        | 409 B      |
| C11                                | 11 | 5,010,456      | 7.233983   | 1,444        | 372 B      |
| C12                                | 12 | 54,721,224     | 76.344777  | <b>1,395</b> | 341 B      |
| <i>Large-Scale Grid Structures</i> |    |                |            |              |            |
| grid 4×4                           | 16 | 4,613,504      | 5.981606   | <b>1,297</b> | 512 B      |
| grid 5×5                           | 25 | 81,920,000     | 171.683820 | <b>2,096</b> | 655 B      |
| <i>Complex Graphs</i>              |    |                |            |              |            |
| complex                            | 13 | 1,282,136      | 2.130338   | 1,662        | 315 B      |
| graph V13F4                        | 13 | 4,236,574      | 6.324251   | 1,493        | 315 B      |
| graph V13F4 v2                     | 13 | 10,225,274     | 16.541969  | 1,618        | 630 B      |
| graph V15F5                        | 15 | 89,897,864     | 146.879791 | 1,634        | 546 B      |
| <i>Medium Graphs</i>               |    |                |            |              |            |
| graph V11F3                        | 11 | 672,902        | 1.025228   | 1,524        | 372 B      |
| graph V9E10F3                      | 9  | 13,880         | 0.026242   | 1,891        | 455 B      |
| graph V12E25F12                    | 12 | 62             | 0.000080   | 1,297        | 682 B      |

Table 5: Additional Test Cases (Table 5)

| Test Case            | V  | Triangulations | Time (s) | ns/Triang | Mem/Vertex |
|----------------------|----|----------------|----------|-----------|------------|
| graph V8E10F4        | 8  | 40             | 0.000353 | 8,817     | 512 B      |
| graph V8E15F8        | 8  | 5              | 0.000010 | 1,954     | 512 B      |
| graph V7E9F3         | 7  | 9              | 0.000016 | 1,764     | 585 B      |
| graph V7E11F5        | 7  | 7              | 0.000013 | 1,798     | 585 B      |
| graph V6E7F3_1       | 6  | 20             | 0.000049 | 2,462     | 682 B      |
| graph V6E7F3_2       | 6  | 24             | 0.000058 | 2,432     | 682 B      |
| graph V6E7F3_3       | 6  | 20             | 0.000039 | 1,933     | 682 B      |
| graph V6E7F3_4       | 6  | 20             | 0.000050 | 2,496     | —          |
| quadrilateral 4quad  | 9  | 16             | 0.000048 | 2,993     | 455 B      |
| quadrilateral 7quad  | 9  | 72             | 0.000201 | 2,793     | 455 B      |
| quadrilateral 14quad | 16 | 16,384         | 0.036991 | 2,258     | 512 B      |
| square7              | 9  | 36             | 0.000102 | 2,844     | 910 B      |
| square8              | 10 | 48             | 0.000179 | 3,730     | 819 B      |
| simple3              | 5  | 4              | 0.000010 | 2,603     | 819 B      |
| W5 wheel             | 6  | 5              | 0.000008 | 1,632     | 682 B      |
| square w/ pentagon   | 5  | 4              | 0.000009 | 2,325     | —          |
| triple rectangle     | 5  | 4              | 0.000011 | 2,661     | 819 B      |

## 7 Conclusions

We have presented the first algorithm for generating all triangulations of biconnected plane graphs with constant amortized time complexity. Our algorithm extends the elegant tree-of-triangulations framework of Parvez, Rahman, and Nakano [11] from triconnected to biconnected graphs by introducing safe root selection and local genealogical trees.

### 7.1 Key Achievements

1. **Theoretical Foundation:** We have provided rigorous proofs establishing:
  - Existence of at least two safe root vertices in every face (Theorem 3)
  - Completeness of the algorithm without duplications (Theorem 2)
  - Amortized  $O(1)$  time complexity per triangulation (Theorem 6)
  - Linear  $O(n)$  space complexity (Theorem 8)
2. **Practical Algorithm:** Efficient implementation suitable for real-world applications with extensive experimental validation
3. **Novel Techniques:** Safe root selection and global chord management strategies that can extend to other enumeration problems involving plane graph decompositions
4. **Broader Applicability:** Extension from triconnected to the more general class of biconnected plane graphs, significantly expanding the scope of graphs that can be processed efficiently

### 7.2 Theoretical Significance

This work makes several theoretical contributions:

- **Generalization:** We show that constant-time triangulation enumeration is achievable beyond the restrictive triconnected case
- **Multi-edge Resolution:** Our safe root concept provides a general strategy for handling chord conflicts in face-decomposed plane graphs
- **Amortized Analysis:** The detailed amortized analysis demonstrates that building time is dominated by output size even with complex face interactions

Each of these problems involves selecting “pivot” vertices or edges to avoid conflicts during recursive decomposition, suggesting that safe root selection strategies may have broad applicability in computational geometry.

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